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Context-aware causal reasoning for explanatory visual question answering

Jiali Miao ^{a,b}, Xiaoling Huang ^c, Kui Yu ^{a,b,*}, Yecheng Shi ^{a,b}, Xiang Wang ^d, Fuyuan Cao ^e^a Key Laboratory of Knowledge Engineering with Big Data (the Ministry of Education of China), Hefei University of Technology, Hefei, 230601, China^b School of Computer Science and Information Engineering, Hefei University of Technology, Hefei, 230601, China^c The School of Computer and Information Engineering, Chuzhou University, Chuzhou, 239000, China^d School of Meteorology and Oceanography, National University of Defense Technology, Changsha, 410073, China^e School of Computer and Information Technology (Big Data College), Shanxi University, Taiyuan, 030006, China

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ABSTRACT

Explanatory Visual Question Answering (EVQA) is a multimodal reasoning task that answers visually relevant natural language questions and generates user-friendly multimodal explanations. Although current EVQA methods can generate grammatically sound and vocabulary appropriate explanations, they still face the challenge of explanatory illusion (i.e., wrong explanations arrive at correct answers). To address this issue, we propose a novel Context-Aware Causal Reasoning (CACR) algorithm. Specifically, we first leverage a Multimodal Large Language Model and design prompting strategies to generate high-quality image descriptions as complementary evidence for helping generate correct explanations. Then, we propose a Structural Causal Model (SCM) to establish the causal relationship among the description, the explanation, and the answer. Finally, we transform the SCM into a deep variational inference network framework that enables causal reasoning to ensure that the explanations are grounded in the visual evidence related to the answer. Extensive experiments show that CACR significantly reduces the risk of explanatory illusion compared to six state-of-the-art methods, outperforming the best baseline by 2.22% on average in terms of explanation quality.

1. Introduction

Visual question answering (VQA) (Anderson et al., 2018; Antol et al., 2015; Yu et al., 2019) aims to answer vision-related language questions. In contrast, Explanatory VQA (EVQA) not only generates direct answers, but also provides a user-friendly explanation to illustrate the reasoning process (Chen & Zhao, 2022). Explanations provide evidential support for answers to reduce user uncertainty. EVQA bridges the gap between visual understanding (Kim et al., 2021; Xie et al., 2024) and natural language reasoning (Devlin et al., 2018) in interpretable AI systems (Ye et al., 2023).

While Multimodal Large Language Models (MLLMs) (Yan et al., 2023; Ye et al., 2023) are increasingly employed in this domain, they face significant challenges in generating satisfactory explanations. Specifically, these models struggle to explain how they arrive at answers. Even when explicitly prompted to show their reasoning process, they often describe images instead. Simply making models bigger or tweaking prompts does not fix the core limitations: the provided explanations are not consistently trustworthy or user-friendly. Therefore, it is crucial to explore alternative MLLM-independent methods for EVQA

to establish causality between answers and explanations and control the reasoning process.

Existing EVQA methods can be categorized into afterward and during-inference explanatory models. The afterward models treat answer prediction and explanation generation as two separate processes, and ignore their connection (Chen & Zhao, 2022; Li et al., 2018). To enable explanations to influence the answer inference process, Xue et al. (2023) integrated the multimodal framework (Li et al., 2019; Tan & Bansal, 2019) for causal reasoning with the variational inference network. They proposed the variational causal inference network (VCIN) as the first work for the during-inference models, aiming to address the consistency problem of answers and explanations.

However, since the model reasoning process does not force explanations to be consistent with visual evidence, the explanations generated by VCIN are often detached from the actual visual content, resulting in illusory explanations. Explanatory illusion is a phenomenon in which the explanation provided by a model's output is incorrect or invalid, yet coincidentally yields the correct answer due to spurious correlation. This is a relational issue between the explanation and the answer, rather than an independent property of either one. As illustrated in Fig. 1, although

* Corresponding author at: Hefei University of Technology, Hefei, 230601, China.

E-mail addresses: yukui@hfut.edu.cn (K. Yu), xiangwangcn@nudt.edu.cn (X. Wang).

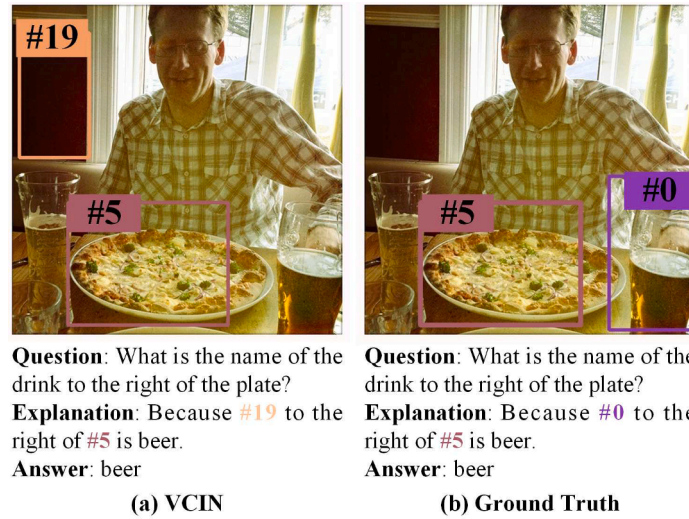


Fig. 1. An example of explanation and answer predicted by VCIN that the wrong explanation gets the right answer.

VCIN provides the correct answer, the visual relationship is incorrect in the generated explanations. Thus, current models lack an effective mechanism to ensure the correctness of key visual evidence for supporting correct explanation generation, resulting in a mismatch between the explanation and the visual facts.

From the visual understanding perspective, all credible evidence for VQA should exist in the image. We can assume that if there exists an image description that provides complete coverage of the multimodal semantics of a visual context (objects, attributes, relationships), all credible evidence for explanations should exist in that description. The reasonable use of image descriptions to enhance the reliability of explanations appears to be an option. The application of image descriptions within EVQA models is not a recent development. Early works, such as VQA-E (Li et al., 2018), explored the use of image descriptions as afterthought explanations for question-answering, but the description itself bears no relation to the question-answering process. As reasoning chains (Qiu et al., 2025; Zhang et al., 2025) gain increasing prominence in VQA, enabling models to extract reasoning evidence from data has become a pivotal concern in EVQA. Thus, we can introduce image description as a global visual context information to provide more evidence for explanations.

Nevertheless, in practical application, obtaining an ideal image description is no simple matter. The generation of high-quality descriptions heavily relies on having sufficient prior knowledge (Pi et al., 2024). Traditional methods for this purpose are often costly and error-prone. The powerful MLLM can provide information-rich descriptions, yet suffers from a significant number of hallucinations that cannot be quantified. Consequently, the lack of high-quality descriptions limits the model's ability to obtain reliable evidence. Thus, we need to address **Challenge 1**: how to obtain high-quality descriptions to enhance the presentation of explanatory evidence? Moreover, in existing EVQA methods, image descriptions are mainly used to generate answers as a supplement to the question stem (Praninick et al., 2025), ignoring the visual evidence support of the descriptions for generating explanations. The explanations thus generated lack reliable visual evidence. Therefore, we need to address **Challenge 2**: how to integrate image descriptions into the reasoning process such that explanations are from their corresponding visual evidence?

To tackle the above challenges, we propose a novel Context-Aware Causal Reasoning (CACR) framework for EVQA. Our main contributions are as follows:

- For Challenge 1, we design an effective prompting strategy to generate high-quality image descriptions by leveraging the frozen MLLM and take them as static complementary evidence base for explanation reasoning.
- For Challenge 2, we first devise a new SCM to establish the causal correlation among descriptions, explanations, and answers. Then, we propose a deep variational inference network framework that enables the joint inference of the description and explanation to ensure the explanations are from their corresponding visual evidence.

Extensive experiments show that CACR significantly reduces the risk of explanatory illusion compared to six state-of-the-art models, outperforming the best baseline by 2.22% on average in terms of explanation quality. To our knowledge, while there have been relevant VQA studies on the inferential form of evidence chains (Qiu et al., 2025; Zhang et al., 2025; Zhao et al., 2024), we are the first to develop a during-inference EVQA model for reasoning that extracts explanatory evidence from image descriptions.

2. Related work

Unlike the traditional VQA methods (Anderson et al., 2018; Hudson & Manning, 2019) which infer answers via black-box processes (Chowdhury & Soni, 2025), EVQA emphasizes the explainability of the questions and answers. Specifically, EVQA focuses on the ability to justify answers through multimodal explanations, and the ability to justify the answers with multimodal (Chen & Zhao, 2022) explanations.

Based on the different roles of multimodal explanations, the existing EVQA methods can be divided into two types: afterward and during-inference explanatory models. As a representative of afterward explanatory models, Li et al. (2018) proposed the Visual Question Answering with Explanation task and constructed the VQAE dataset, which is based on the VQA 2.0 dataset (Antol et al., 2015) and MSCOCO image descriptions. After that, Chen and Zhao (2022) were the first to define the EVQA task, introducing a novel form of multimodal explanation by developing a functional program to construct explanations automatically. They established the GQA-REX dataset based on the GQA (Hudson & Manning, 2019) dataset. Li et al. (2024) proposed CRVQA, which both generates textual descriptions and object bounding boxes.

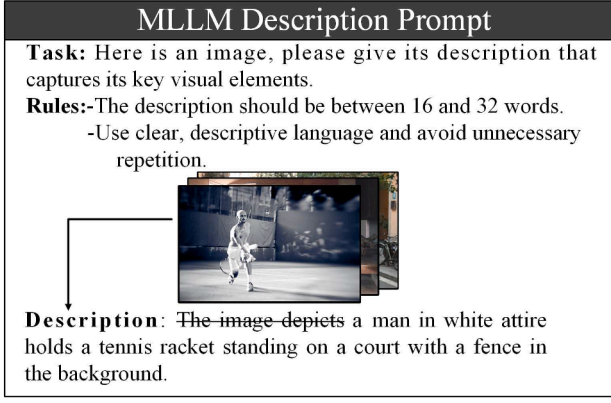


Fig. 2. MLLM prompt.

To bridge the gap between explanation and answering inference processes, Xue et al. (2023) proposed the VCIN (Variational Causal Inference Network) as an initial study for the during-inference models. They approached the EVQA task from a causal perspective (Chen et al., 2020) by analyzing the underlying Structural Causal Model (SCM), which improves the credibility and quality of the inferred results in the EVQA task. With the development of MLLMs, Xue et al. (2024b) introduced MEAgent, which can generate multimodal explanations with few-shot training. However, existing EVQA studies directly generate natural language explanations from visual multimodal features (Xue et al., 2024a,c), overlooking the capacity of global semantic information within image descriptions to support the attribution of evidence sources. The few models (Li et al., 2018) that exist merely employ descriptions as the afterward explanations and supplementary information. Our work extends existing SCM framework (Xue et al., 2023) by introducing image descriptions as a source of contextual evidence for EVQA inference; its theoretical foundation and causal assumptions are detailed in Section 4.2.

3. Notations

EVQA aims to answer a given question with a relevant image and generate a multimodal explanation of the reasoning process. The input can be denoted as (Q, I) . $Q = q_1 q_2 \dots q_m$ is a textual question of total m words and q_i is the i th word in the question. The image I can be represented by an RGB tensor. The output of an EVQA model can be denoted as (A, E, C) . A is the answer selected from a predefined dictionary set $\{a_1, \dots, a_U\}$ of total U answers. $E = e_1 e_2 \dots e_n$ and $C = c_1 c_2 \dots c_m$ are the explanation and description of the total n and m tokens, where C is the pre-extract image descriptions as the global visual context, e_i and c_j are the i th and j th token that from a predefined vocabulary or an index number for the visual region. The ground truth output (A', E', C') is given only in the training process.

4. The proposed CACR method

In this section, we introduce our description generation process and a novel SCM based on deep variational inference within the CACR framework. Section 4.1 details the method for pre-extracting image descriptions using MLLM prompts. Section 4.2 introduces the proposed SCM, followed by Sections 4.3 and 4.4, which demonstrate its implementation based on deep variational inference. Sections 4.5–4.7 describe the neural network modules that realize causal paths, including the context generator, explanation generator, multimodal encoder, and answer classifier.

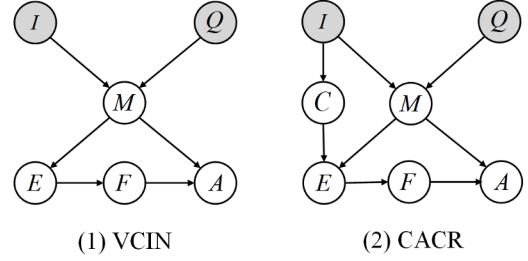


Fig. 3. Comparison of structural causal models VCIN and CACR for EVQA, I and Q denote the image and question, M denotes the pre-trained multimodal features, C denotes the image description that presents the context of visual question, E denotes the explanation generated from M and C , F denotes the robust feature representation of explanation, A denotes the answer.

4.1. Pre-extracted description

To obtain stable, trainable image description data containing sufficient explanatory evidence, we designed an MLLM prompting method to pre-extract visual image descriptions of problems. We choose Qwen2-VL (Wang et al., 2024) as a sample model to ensure the basic quality of the descriptions. This strategy requires suppressing the generation of hallucinatory descriptions while preserving MLLM's ability to produce semantically rich outputs. When utilizing MLLM to pre-extract descriptions, we need to design a prompting strategy to obtain high-quality image descriptions. This strategy requires suppressing the generation of hallucinatory descriptions while preserving MLLM's ability to produce semantically rich outputs. Thus, we need to tackle the following three issues: 1) The comprehensiveness of visual elements: the description needs to fully capture all the key objects in the image and their spatial relationships. 2) The accuracy of the description: the description needs to avoid omitting important salient objects while preventing irrelevant details. 3) The computational efficiency of generation: embedding the MLLM into model training requires a lot of GPU memory and computational power.

The specific prompt design is illustrated in Fig. 2. First, we clarify the task goals through prompts to ensure visual coverage of the descriptions, so we ask the MLLM to give the image description that captures key visual elements. Then, the output length is limited by adding rule prompts (between 16 and 32) and requiring the output description to avoid unnecessary repetitive information, to prevent irrelevant details. Finally, by pre-extracting the offline description set of the dataset, experiments and validations are performed in a static environment to save computational power. In addition, we process descriptions by removing independent starters such as "the image shows". The descriptions are denoted as $C = c_1 c_2 \dots c_m$. In the experiments, we employed causal intervention experiments to replace MLLM descriptions, thereby validating the dependency of CACR on MLLM performance and description quality. Moreover, due to the inability to quantify the degree of hallucination, we also present a visual comparison of MLLM outputs before and after prompting to demonstrate how prompts suppress descriptive hallucinations.

4.2. The improvement of structural causal model

SCM is a formal framework for representing causal relationships between variables and data-generating mechanisms in the inference process. As illustrated in Fig. 3(1), VCIN (Xue et al., 2023) introduces the robust feature F in SCM, which allows explanations to engage in the model reasoning process. However, the image semantics carried in M , which generate E in VCIN, are not clear enough to point to verifiable evidence. And the SCM path does not force E to align with visual evidence, leading to explanations that are detached from the actual visual content.

To complement the reliable visual evidence in the explanation, we introduce the image description as the global visual context C into the reasoning process. Compared to using only direct visual regional features extracted from object detectors, image descriptions provide a more comprehensive and structured semantic summary of visual scenes. Regional features are often sparse, incomplete, or noisy, as detectors may overlook small objects, generate inaccurate bounding boxes, or fail to capture fine-grained attributes and relationships between objects. In contrast, when generated appropriately, descriptions can explicitly encode global context, combinations of attributes, and relational information in a compact textual form, which naturally aligns with the language-based reasoning required by EVQA. Furthermore, descriptions are less affected by the domain gap between pre-training on detection and VQA inference, and they can incorporate high-level prior knowledge. However, descriptions are not entirely error-free; if the prompting strategy is not optimized, they may introduce irrelevant details or omit key objects. Therefore, we treat descriptions as supplementary evidence to visual features rather than a replacement, and integrate them via causal paths, enabling the model to leverage beneficial information while remaining robust to potential noise.

As illustrated in Fig. 3(2), our SCM is an extension of the VCIN framework. It follows the Deep Structural Causal Model (DSCM) (Pawlowski et al., 2020) paradigm, in which SCM serves as an inference abstraction to guide the design of neural modules, rather than a data-generation process that must be identified from observational data. In the implementation, the VLPN encodes I and Q into M , while a separate context generator generates C from I via an independent path. These two paths are implemented by distinct modules with independent parameters. The VLPN's encoding is completed before the context generator runs, meaning that the path from C back to M via I is physically disconnected in the data flow. This modular separation prevents backdoor paths from C to E via M . Thus, when conducting a causal sensitivity analysis of $C \rightarrow E$ through intervention experiments, we ignore the SCM path from (I, Q) to M , as detailed in Section 5.9.

Specifically, we first add a new path $I \rightarrow C$ in the SCM, as illustrated in Fig. 3(2), to represent the description generation process. Then we introduce the path $C \rightarrow E \leftarrow M$, which realizes the descriptions as visual context information to generate the explanation with the multimodal features. Thus, the path $E \rightarrow F$, which realizes the explanations on deep variational inference, where F is the robust feature derived from the explanation that contains information from C to enhance its representation. Finally, across the path $F \rightarrow A \leftarrow M$, we facilitate the conversion of the structure to deep variational inference.

We model F to a gaussian distribution to improve the robustness of F that $F \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$, where $\{\mu, \sigma^2\}$ are d_f -dimensional vectors computed from E and C . Moreover, to mitigate the evidence bias between E and C , we minimize the Kullback-Leibler (KL) divergence of $KL(q(F | E^*) \parallel q(F^c | C^*))$, where E^*, C^* are the predicted explanation and visual context, and F, F^c are their robust features. The latter ensures that the semantic distribution of the explanation aligns with that of the visual context, while the former ensures that this alignment is effectively utilized during the inference process. This KL-based evidence constraint acts as an inductive bias, reducing the risk of explanation illusions by penalizing significant semantic discrepancies between the explanation and the visual context. We emphasize that this latent-space alignment is not a proof of regional-level visual faithfulness, but rather a step toward it by promoting semantic coherence, thereby laying the foundation for downstream assessments of the explanation's credibility.

4.3. Deep variational inference

To implement our SCM, the first objective of the training is to maximize the Evidence Lower Bound (ELBO) (Yang, 2017) of the marginal likelihood (Kingma et al., 2013) of predicting the true answer A' while

modeling all causal effects to it. This objective represents the path of $F \rightarrow A \leftarrow M$ that we keep in mind with the VCIN (Xue et al., 2023) in our theoretical reasoning, as follows:

$$\begin{aligned} & \log p(A' | M) \\ & \geq E_{q(F|M)} [\log p(A' | M, F) + \log p(F | M) - \log q(F | M)] \\ & = E_{q(F|M)} [\log p(A' | M, F)] - KL(q(F | M) \parallel p(F | M)) \\ & = E_{q(F|E')} [\log p(A' | M, F)] - KL(q(F | E') \parallel p(F | M)), \end{aligned} \quad (1)$$

where p and q are the distributions of our training. The expectations follow the lemma (Xue et al., 2023):

$$q(F | M) = \sum_E q(F | E)q(E | M) = q(F | E'), \quad (2)$$

since $q(F | M)$ is a Dirac delta distribution satisfying $q(E' | M) = 1$. To allow explanations to take evidence from the description C , we introduce the description C as visual context into the SCM as the path of $M \rightarrow E \leftarrow C$. Thus, all relevant variables for generating A do not depend on M alone, and its ancestor nodes include C . This will result in the learning objective of CACR changing to $\log p(A' | M, C)$, and $q(E' | M) = 1$ unsatisfied, and $q(E' | M, C) = 1$ satisfied. Thus, the lemma in Eq. (2) of CACR is reformulated as follows:

$$q(F | M, C) = \sum_E q(F | E)q(E | M, C) = q(F | E'). \quad (3)$$

Proof. Let $I(X; Y)$ denote the mutual information. By the SCM defined in Fig. 3(2), the causal paths $M \rightarrow E \leftarrow C$ and $E \rightarrow F$ establish the chain $(M, C) \rightarrow E \rightarrow F$. By the Markov property of this chain, F is conditionally independent of (M, C) given E :

$$p(F | M, C, E) = p(F | E). \quad (4)$$

Since E is a deterministic function of (M, C) produced by the generator, the conditional distribution $p(F | M, C)$ is a degenerate distribution that places all probability mass on the single value $E = f(M, C)$, i.e., $p(E | M, C) = 1$ at $E = f(M, C)$ and 0 otherwise. Expanding $p(F | M, C)$ by marginalizing over E :

$$p(F | M, C) = \int p(F | M, C, E) p(E | M, C) dE. \quad (5)$$

Because $p(F | M, C)$ is degenerate at $E = f(M, C)$, this integral evaluates to $p(F | M, C, E)$ with E fixed to its deterministic value. Combining this with the conditional independence in Eq. (4) gives:

$$p(F | M, C) = p(F | M, C, E) = p(F | E). \quad (6)$$

This establishes that E is a sufficient statistic for F with respect to (M, C) . With this condition established:

- For the posterior, $p(F | E') = p(F | M, C)$ follows directly from Eq. (6). Since $q(F | E')$ is the variational approximation to the true posterior, it approximates $p(F | E')$, yielding $q(F | E') \approx p(F | M, C)$;
- For the prior at test time, E^* is the generated explanation. Under the same sufficiency condition, $p(F | E^*) = p(F | M, C)$.

Thus, both approximations are direct consequences of the Markov property of the SCM structure. The training objectives introduced subsequently are designed to drive the network toward this sufficiency condition.

Due to the pre-extraction process for the description, the learning objective of CACR is represented as:

$$\begin{aligned} & \log p(A' | M, C) \\ & \geq E_{q(F|M,C)} [\log p(A' | M, C, F) + \log p(F | M, C) - \log q(F | M, C)] \\ & = E_{q(F|E')} [\log p(A' | M, C, F)] - KL(q(F | E') \parallel p(F | M, C)), \end{aligned} \quad (7)$$

However, $p(F | M, C)$ is difficult to compute because its complexity grows exponentially with the number of variables in M and C . We propose to utilize an approximation $p(F | E^*)$ that corresponds to the model test where E^* is the generated explanation. In this way, we realize the

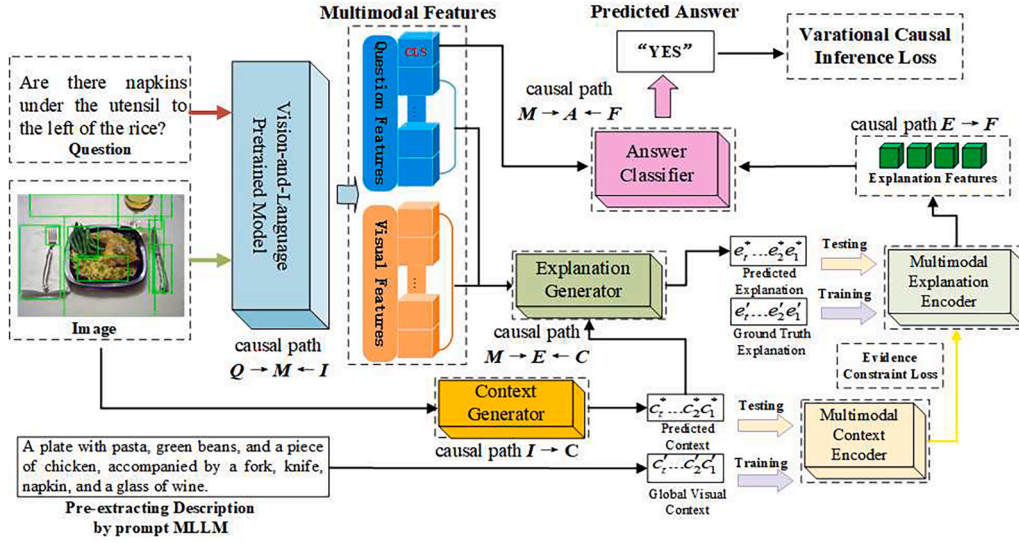


Fig. 4. The framework of CACR: (1) causal path $Q \rightarrow M \leftarrow I$ is implemented by a VLP, which generates the multimodal feature consisting of question features and visual features; (2) causal path $I \rightarrow C$ is implemented by the context generator and the description is generated; (3) causal path $M \rightarrow E \leftarrow C$ is implemented by the explanation generator and the explanation is generated; (4) Causal path $E \rightarrow F$ is the multimodal explanation encoder and the multimodal context encoder to generate the robust explanation feature and calculate the evidence constraint loss; (5) Causal path $M \rightarrow A \leftarrow F$ is implemented by the answer classifier and the answer is predicted.

computability of $p(F | M, C)$ by bringing the high-dimensional M and C to generate the low-dimensional E^* into Eq. (7). Thus, we get the variational causal inference loss:

$$\begin{aligned} \mathcal{L}_{ans} &= -E_{q(F|E^*)} [\log p(A' | M, C, F)] + KL(q(F | E^*) \parallel p(F | E^*)) \\ &= -E_{q(F|E^*)} [\log p(A' | M, C, F)] \\ &\quad + \frac{1}{2} \left[\log \frac{|\Sigma_{(E^*)}|}{|\Sigma_{(E')}|} - d_f + \text{tr}\{\Sigma_{(E^*)}^{-1} \Sigma_{(E')} \} + \Delta \mu^T \Sigma_{(E^*)}^{-1} \Delta \mu \right], \end{aligned} \quad (8)$$

where $\Sigma_{(E^*)}^{-1} = \text{diag}(\sigma_{(E^*)}^2)$, $\Sigma_{(E')} = \text{diag}(\sigma_{(E')}^2)$, $\Delta \mu = (\mu_{(E^*)} - \mu_{(E')})$, $(\mu_{(E^*)}, \sigma_{(E^*)}^2)$ and $(\mu_{(E')}, \sigma_{(E')}^2)$ are demoted as d_f -dimensional vectors computed from F .

Besides maximizing $\log P(A' | M, C)$ of predicting the true answer A' , we also aim at maximizing $p(E' | M, C)$ of the ground truth explanation E' by using the following loss:

$$\mathcal{L}_{exp} = -\log p(E' | M, C). \quad (9)$$

We still aim at maximizing $p(C' | I)$ of the global visual context C' by using the following loss:

$$\mathcal{L}_{con} = -\log p(C' | I). \quad (10)$$

Moreover, to constrain the presence of evidence for generating explanations in the visual context information, we devise an evidence constraint loss:

$$\begin{aligned} \mathcal{L}_{evi} &= KL(q(F | E^*) \parallel q(F^c | C^*)) \\ &= \frac{1}{2} \left[\log \frac{|\Sigma_{(E^*)}|}{|\Sigma_{(C^*)}|} - d_f + \text{tr}\{\Sigma_{(E^*)}^{-1} \Sigma_{(C^*)} \} + \Delta \mu^T \Sigma_{(E^*)}^{-1} \Delta \mu \right] \end{aligned} \quad (11)$$

where F^c is the robust visual context feature that is generated like F and follows gaussian distribution, $F^c \sim \mathcal{N}(\mu_c, \text{diag}(\sigma_c^2))$, and $\Sigma_{(C^*)} = \text{diag}(\sigma_{(C^*)}^2)$, $\Delta \mu = (\mu_{(E^*)} - \mu_{(C^*)})$. This loss can minimize the semantic bias between the generated explanation and visual context through KL scatter to achieve an effective constraint on the semantic reliability of the explanation evidence. This constraint operates at the semantic distribution level and does not require explicit regional-level supervision. It ensures that explanations remain within the semantic space defined

by the visual context, thereby reducing the likelihood of generating explanations that appear linguistically plausible without visual evidence. By optimizing $\mathcal{L}_{ans}$, $\mathcal{L}_{exp}$, $\mathcal{L}_{con}$, and $\mathcal{L}_{evi}$, we can train our SCM to predict accurate answers and generate visual context C to enhance generating rational explanations. Next, we will introduce the implementation of the SCM into a deep variational inference network as shown in Fig. 4.

4.4. Multimodal feature encoder

In this subsection, we introduce the process of multimodal feature representation to implement path $Q \rightarrow M \leftarrow I$ in our SCM. We adopt LXMERT (Tan & Bansal, 2019) as the Vision-and-Language Pretrained Model (Li et al., 2022a,b; Lu et al., 2019) (VLP) in our experiments. The input to the LXMERT comprises two modalities: (1) a set of 36 visual objects $\{(f_i, p_i)\}_{i=1}^{36}$ extracted from image I using Faster R-CNN (Ren et al., 2015), where $f_i \in \mathbb{R}^{2048}$ represents the region-of-interest (ROI) feature and $p_i \in \mathbb{R}^4$ denotes the position vector of the i th object; (2) a question sequence formatted as $q_0 q_1 \dots q_m q_{m+1}$, where $q_0 = [CLS]$ and $q_{m+1} = [EOS]$ are appended to the original textual input. Then the multimodal features $M = \{V, T\}$ as follows:

$$V, T = \text{LXMERT}(\{(f_i, p_i)\}_{i=1}^{36}, q_0 q_1 \dots q_m q_{m+1}), \quad (12)$$

where $V \in \mathbb{R}^{36 \times 768}$ are visual features of 36 regions and $T \in \mathbb{R}^{(m+2) \times 768}$ are question features of $(m+2)$ tokens.

4.5. Generators of explanation and context

Although we have constructed an offline set of descriptions to serve as the foundation for visual evidence in explanations, directly incorporating these descriptions into the model leads to information leakage. Specifically, the model would directly obtain high-level semantic summaries of images during training, thereby effectively bypassing the need to learn visual feature extraction and multimodal alignment from the raw visual input. This shortcut causes the model to rely on pre-extracted descriptions rather than learning to reason from low-level visual features, severely compromising its generalization ability. To avoid this issue, we propose a context generator that converts visual features into coherent contextual text. This module is trained using a context generation loss $\mathcal{L}_{con}$, with offline MLLM descriptions serving as ground truth.

This ensures that the model learns to generate descriptions from visual inputs rather than directly exploiting them as privileged information.

So we implement path $I \rightarrow C$ to introduce the description information by training a generator. After training, the model can learn the descriptions by itself to generate reasoning explanations without having to draw from the evidence base. Thus, we propose a context generator that translates the visual features into a coherent contextual text by constraining Eq. (10). When generating the t th token of the description, we first obtain token embeddings of previous $(t-1)$ generated tokens $\{c_i\}_{i=1}^{t-1}$ as follows:

$$\text{emb}(c_i) = \text{WordEmb}_c(c_i), \quad (13)$$

where $\text{WordEmb}_c(\cdot)$ is the description embedding function. Denote $\mathbf{G} = [\text{emb}(c_1), \dots, \text{emb}(c_{t-1})] \in \mathbb{R}^{(t-1) \times 2048}$, we then fuse the extracted image features through a L_c -layer Transformer (Vaswani et al., 2017). The fusion process as:

$$\begin{aligned} \mathbf{G}^0 &= \mathbf{G}, \\ \bar{\mathbf{G}}^l &= \text{TransLayer}^l(\mathbf{G}^{l-1}), \\ \mathbf{G}^l &= \text{CrossLayer}^l(\bar{\mathbf{G}}^l, \mathbf{I}, \mathbf{I}), \\ l &= 1, 2, \dots, L_c, \end{aligned} \quad (14)$$

where $\text{TransLayer}^l(\cdot)$ is the l th self-attention layer and $\text{CrossLayer}^l(\cdot, \cdot, \cdot)$ is the l th cross-attention layer. Then, we employ an MLP to predict the word probabilities over the text vocabulary:

$$p^l = \text{softmax}(\text{LN}(\text{GELU}(\mathbf{G}_{t-1}^{L_c} \mathbf{W}_c^l)) \mathbf{W}_c^2) \in \mathbb{R}^K, \quad (15)$$

where $\mathbf{W}_c^1 \in \mathbb{R}^{2048 \times d_c}$ and $\mathbf{W}_c^2 \in \mathbb{R}^{d_c \times K}$ are learnable parameters, $\mathbf{G}_{t-1}^{L_c}$ is the output feature corresponding to the $(t-1)$ th token from the L_c th Transformer layer, and K is the size of the vocabulary. The t th token is then generated by selecting the token with the highest probability:

$$c_t^* = \underset{i}{\text{argmax}} p_i^l. \quad (16)$$

We engage C in the process of generating E by path $C \rightarrow E$, while path $M \rightarrow E$ preserves the multimodal explanation output. Thus, we implement path $M \rightarrow E \leftarrow C$ in our SCM and constraints Eq. (9) to generate more credible explanations. We adopt the multimodal explanation gating transformer (Wu & Mooney, 2018) proposed in VCIN as the explanation generator, and connect M and the output of the context generator $C^* = c_1^* \dots c_t^*$ as generator input. Especially, the explanation generator outputs both text and visual token indexes. When generating the t th token in the explanation, we compute token embeddings of previous $(t-1)$ generated tokens $\{e_i\}_{i=1}^{t-1}$ as follows:

$$\text{emb}(e_i) = \begin{cases} \text{WordEmb}_e(e_i), & \text{if } e_i \text{ is a word token,} \\ \mathbf{V}_j, & \text{if } e_i = \#j \text{ is a visual token,} \end{cases} \quad (17)$$

where $\text{WordEmb}_e(\cdot)$ is the explanation embedding function. Denote $\mathbf{B} = [\text{emb}(e_1), \dots, \text{emb}(e_{t-1})] \in \mathbb{R}^{(t-1) \times 768}$, then the fusion process is formulated as:

$$\begin{aligned} \mathbf{C} &= \text{linear}(\mathbf{C}^*) \\ \mathbf{B}^0 &= \mathbf{B}, \\ \bar{\mathbf{B}}^l &= \text{TransLayer}^l(\mathbf{B}^{l-1}), \\ \mathbf{B}^l &= \text{CrossLayer}^l(\bar{\mathbf{B}}^l, [\mathbf{V}, \mathbf{T}, \mathbf{C}], [\mathbf{V}, \mathbf{T}, \mathbf{C}]), \\ l &= 1, 2, \dots, L_e, \end{aligned} \quad (18)$$

where linear is a linear layer for dimension transformation.

4.6. Multimodal encoders of explanation and context

To realize a variational inference process for the explanation of answers, we implement path $E \rightarrow F$ in our SCM and derive the representation of the robust feature F . We first insert a $[CLS]$ token to both the explanation text sequences, followed by a Transformer to learn the robust features:

$$\begin{aligned} \theta &= [\text{emb}([CLS], \text{emb}(e_1), \dots, \text{emb}(e_T))] \in \mathbb{R}^{(T+1) \times 768}, \\ \mathbf{F} &= \text{Transformer}_e(\theta) \in \mathbb{R}^{(T+1) \times 768}. \end{aligned} \quad (19)$$

To calculate the evidence constraint loss, we need to derive the robust context feature $F^c = F(C)$:

$$\begin{aligned} \mathbf{O} &= [\text{emb}([CLS], \text{emb}(c_1), \dots, \text{emb}(c_S))] \in \mathbb{R}^{(S+1) \times 2048}, \\ \mathbf{F}^C &= \text{Transformer}_c(\mathbf{O}) \in \mathbb{R}^{(S+1) \times 2048}, \end{aligned} \quad (20)$$

where we employ the embedding function referenced in Eqs. (13) and (17). θ and $\mathbf{O}$ are token embeddings of the generated explanation and context, $\text{Transformer}_e(\cdot)$ and $\text{Transformer}_c(\cdot)$ denote the 4-layer and 8-layer Transformer modules, T and S are the lengths of the explanation and visual context. We also utilize the semantic information of context features to complement the global semantic consistency of explanation features:

$$\mathbf{F} = [\mathbf{F}_0 \mid \mathbf{F}_0^C], \quad (21)$$

where $\mathbf{F}_0$ and $\mathbf{F}_0^C$ are features of $[CLS]$, and $[\cdot \mid \cdot]$ is the concatenation function. Assuming $F \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$, we can get the gaussian parameters by a 2-layer MLP:

$$\begin{bmatrix} \mu \\ \log \sigma^2 \end{bmatrix} = \text{MLP}(\mathbf{F}). \quad (22)$$

Ultimately, to realize the deep impact of explanation on answer prediction, we implement path $M \rightarrow A \leftarrow F$ to predict the answer A .

4.7. Answer classifier

To realize the deep impact of explanation on answer prediction, we implement path $M \rightarrow A \leftarrow F$ to predict the answer A , we leverage F and the multimodal $[CLS]$ token feature T_0 to compute $P(A \mid M, C, F)$ as follows:

$$\begin{aligned} P(A \mid M, C, F) \\ = \text{softmax}(\text{LN}(\text{GELU}([F \mid T_0] \mathbf{W}_a^1)) \mathbf{W}_a^2) \in \mathbb{R}^U, \end{aligned} \quad (23)$$

where $\mathbf{W}_a^1 \in \mathbb{R}^{1536 \times d_a}$ and $\mathbf{W}_a^2 \in \mathbb{R}^{d_a \times U}$ are learnable parameters, and U represents the total number of possible answers. During the test, to prevent sampling bias and uncertain results, we predict the answer by estimating the expectation of $p(A \mid M, F)$ using the Normalized Weighted Geometric Mean (NWGM) approximation (Xue et al., 2023) as follows:

$$\begin{aligned} E_F\{p(A \mid M, F)\} \\ = E_F\{\text{softmax}(\text{LN}(\text{GELU}([F \mid T_0] \mathbf{W}_a^1)) \mathbf{W}_a^2)\} \\ \approx \text{softmax}(\text{LN}(\text{GELU}([E\{F\} \mid T_0] \mathbf{W}_a^1)) \mathbf{W}_a^2) \\ = \text{softmax}(\text{LN}(\text{GELU}([\mu \mid T_0] \mathbf{W}_a^1)) \mathbf{W}_a^2). \end{aligned} \quad (24)$$

4.8. Optimization

We train our model by optimizing the losses proposed in Eqs. (8)–(10) as follows:

$$\begin{aligned} \mathcal{L}_{ans} \\ \approx -\frac{1}{H} \sum_{i=1}^H A' \log p(A \mid M, C, F_i) \\ + \frac{1}{2} \left[\log \frac{|\Sigma_{(E^*)}|}{|\Sigma_{(E^*)}|} - d_f + \text{tr} \left\{ \sum_{(E^*)}^{-1} \sum_{(E^*)} \right\} + \Delta \mu^T \sum_{(E^*)}^{-1} \Delta \mu \right], \\ \mathcal{L}_{exp} = -\sum_{i=1}^T e'_i \log y^i - \sum_{i=1}^T [\omega'_i \log \omega_i + (1 - \omega'_i) \log (1 - \omega_i)], \\ \mathcal{L}_{con} = -\sum_{i=1}^T c'_i \log p^i, \end{aligned} \quad (25)$$

where A' is the ground truth answer, ω'_i is the ground truth gating value of the i th step, e'_i is the i th ground truth explanation token, y^i is the i th explanation token final prediction probability, c'_i is the i th generated global visual context token, p^i is the i th visual context token final

prediction probability, $F_i \sim \mathcal{N}(\mu_{(E')}, \text{diag}(\sigma_{(E')}^2))$ are H i.i.d. sample. At last, combine Eq. (11), the final loss $\mathcal{L}$ is as follow:

$$\mathcal{L} = \mathcal{L}_{ans} + \mathcal{L}_{exp} + \mathcal{L}_{con} + \mathcal{L}_{evi} \quad (26)$$

5. Experimental evaluation

5.1. Datasets

In our experiments, we have carefully selected a series of authoritative benchmark datasets from the current field of EVQA, based on the type of explanation and the evaluation objective. Common EVQA datasets can be categorised into natural language explanation datasets and multimodal explanation datasets based on the form of their explanations. Since the core of CACR focuses on causal reasoning for multimodal explanations, the multimodal explanation dataset GQA-REX (Chen & Zhao, 2022) naturally becomes our primary dataset for evaluating explanation generation quality. GQA-REX is built upon GQA's (Hudson & Manning, 2019) balanced training, balanced validation, and standard test sets.

To comprehensively validate CACR's performance in answer prediction, we conducted extensive evaluations across four datasets:

- GQA-REX (Chen & Zhao, 2022): Used to verify whether the model maintains high-precision answer prediction capabilities in settings requiring multimodal explanation generation.
- GQA (Hudson & Manning, 2019): Serves as the standard test set to evaluate the model's generalization performance on complex combinatorial reasoning problems.
- GQA-OOD (Kervadec et al., 2021): Examines the model's robustness in out-of-distribution scenarios, testing its resistance to linguistic and visual biases.
- VQAE (Li et al., 2018): Validates whether our model can transfer and generate competitive textual reasoning chains when confronted with pure natural language explanation demands.

Through comprehensive evaluation across these datasets, we aim to demonstrate that our algorithm not only excels in multimodal explanation generation but also maintains robust and powerful visual question-answering performance across diverse challenging scenarios.

5.2. Experimental parameter settings

MLLM defaults to the Qwen2-VL-7B (Wang et al., 2024) model with a parameter count size of 7B. We adopt the BERT Adam (Devlin et al., 2019) optimizer for training our model. For a fair comparison, we used the same optimizer when reproducing VCIN (Xue et al., 2023), which resulted in experimental results that were slightly different from the original paper. The mini-batch size is 128, and the initial learning rate for all trainable parameters is $1e-5$. To mitigate overfitting, we clip gradients with a norm maximum of 1.0. The hidden unit dimensions d_c and d_a are set as 4096 and 1536, respectively. All attention layers employ multi-head attention with 4 heads. The dropout rate of all attention layers is set as 0.1 for regularization. The layer numbers L_c and L_e of the context generator and the explanation generator are set as 8 and 4, respectively. We set the maximum explanation text tokens length T to 18 and the maximum visual contextual text tokens length S to 32. The Monte Carlo sampling number H is set as 4. The entire model is trained on two A5000 GPUs using PyTorch under a Linux operating system.

5.3. Analysis on explanation and image description for attributes

In Table 1, we compare the difference between explanation and image description in terms of modality, dictionary, and format. Since image description focuses on the language modality, it can more accurately recognize and interpret textual content. The dictionary size of

Table 1

Comparison of explanation and image description.

Type	Modality	Dictionary	Format
Explanation	Language-Vision	1839	Customized
Image description	Language	9963	Standard English

image descriptions is notably larger than that of explanations, allowing for a richer and more diverse vocabulary when generating or interpreting image descriptions. Furthermore, the format of image description is standard English, making it more accessible and widely applicable.

5.4. Baseline and evaluation metrics

To evaluate the performance of CACR on the EVQA task, we conduct comparative experiments with several baseline methods.

- VQA-E (Li et al., 2018) constructs an LSTM-based language model that jointly generates the answers and the afterward explanations from MSCOCO image descriptions.
- EXP (Wu & Mooney, 2018) integrates visual tokens into its explanation generator, which is based on LSTM.
- REX (Chen & Zhao, 2022) employs a gating LSTM for generating explanations through multimodal feature fusion.
- LININ (Xue et al., 2024a) restricts the range of candidate answers based on first-order logic to generate explanations.
- VCIN (Xue et al., 2023) is the state-of-the-art method on the leaderboard.

To ensure a fair comparison, we reproduce VCIN on the same optimizer configuration and datasets as our implementation, which yields experimental results that deviate slightly from those reported in the original study. We evaluate the quality of explanation generation by using five language metrics and the visual region metrics Grounding, where BLEU-4 (Papineni et al., 2002) evaluates explanation quality based on gram accuracy, METEOR (Lavie & Agarwal, 2007) evaluates the recall and synonym/stem matching of generated explanations, ROUGE-L (Lin, 2004) evaluates the explanation sentence structure similarity using the longest common subsequence, CIDEr (Vedantam et al., 2015) evaluates the amount of information in the generated explanations using TF-IDF-weighted grams, SPICE (Anderson et al., 2016) evaluates the semantic accuracy of generated explanations based on scene graph matching and Grounding (Chen & Zhao, 2022) evaluates the ability of correctly grounding visual regions in the generated explanations. It is worth noting that our assessment of explanation quality does not rely on any single metric. Following previous research, we evaluate explanations from multiple perspectives: linguistic fluency (BLEU-4, METEOR, ROUGE-L, CIDEr), semantic accuracy (SPICE), and visual grounding ability (Grounding). While our KL-based evidence constraints help promote semantic alignment between the explanation and the visual context, the Grounding metric more directly assesses whether the model can correctly localize the visual regions referenced in the explanation. Together, these metrics provide a multi-perspective evaluation of explanation quality from linguistic, semantic, and visual grounding aspects.

Answering accuracy is measured by calculating Accuracy on all datasets. Moreover, to evaluate the consistency between predicted answers and explanations, we use the automatic metric named Consistency (Con.) in VCIN to compute the rate of explanations that contain the corresponding answers. To align with human judgement, we have extended the explanation-context consistency (CoT.) metric beyond the two evaluation criteria established in VCIN: visual consistency (Vis.) and textual consistency (Tex.), whether the predicted visual and textual tokens in explanations are consistent with the predicted answers and generate Context, respectively. A 5-grade marking system is applied, with 5 as

Table 2

Results of CACR on explanation generation and question answering for EVQA compared with existing methods. REX-VisualBert and REX-LXMERT denote the use of different VLPs. GQA- and OOD- denote answering accuracy on GQA-REX and GQA-OOD datasets. Underlined numbers are sub-optimal results. * is trained on the cleaned datasets. “-” indicates experimental results that could not be retrieved or reproduced.

Model	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	Grounding	GQA-REX-val	GQA-test	OOD-val	OOD-test	VQAE
VQA-E	42.56	34.51	73.59	358.20	40.39	31.29	65.19	57.24	49.20	46.28	61.57
EXP	42.45	34.46	73.51	357.10	40.35	33.52	65.17	56.92	49.43	47.69	-
REX-VisualBert	54.59	39.22	78.56	464.20	46.80	67.95	66.16	57.77	50.26	48.26	68.94
REX-LXMERT	54.79	39.51	79.41	466.01	49.98	70.79	78.19	58.15	71.23	52.15	-
LININ*	58.41	41.45	81.31	515.80	54.49	<u>76.05</u>	81.64	60.59	74.64	54.18	-
VCIN	<u>58.72</u>	<u>41.58</u>	<u>81.37</u>	<u>519.11</u>	<u>54.55</u>	75.83	81.33	59.29	74.49	<u>54.29</u>	<u>69.39</u>
CACR(Ours)	59.21	41.81	81.61	524.90	55.00	77.97	<u>81.59</u>	<u>59.82</u>	78.36	55.04	72.08

Table 3

Results of consistency by automatic metric and human evaluation.

Model	Con.	Vis.	Tex.	CoT.	Average
REX	84.90	2.84	3.24	2.66	2.91
VCIN	90.00	3.32	3.48	2.76	3.19
CACR	93.55	3.64	3.62	3.04	3.43

the maximum grade and 1 as the worst. We randomly select 300 validation samples and employ three professional annotators to conduct a blind evaluation. As annotated explanations are unavailable in the test set, we evaluate the generated explanations only on the validation set of GQA-REX.

5.5. Results and discussion

The experimental results evaluate the quality of explanation by using five language metrics from BLEU-4 to SPICE and the visual regions metric Grounding, as presented in Table 2. CACR achieves the best results in the comparison of methods on uncleaned datasets. All of them show statistically significant differences compared to the suboptimal model, CACR boosts all five language metrics as well as the CIDEr metric most notably, where achieving a 0.49 improvement in BLEU-4, a 0.23 improvement in METEOR, a 0.24 improvement in ROUGE-L, a 5.79 improvement in CIDEr, a 0.45 improvement in SPICE, and 1.92 improvement in Grounding. These results indicate that CACR can generate more reasonable explanations from the description. CACR also enhances accuracy for answers. In particular, we achieve improvements of 3.87 and 0.75 on the OOD-val and OOD-test, which suggests CACR possesses strong robust generalization capability and mitigates the bias inherent in the dataset. Moreover, CACR achieved a 2.69 performance improvement on the VQAE dataset, demonstrating that extracting explanatory evidence from descriptions yields more reliable results than directly using descriptions as explanations. This also validates CACR’s generalisation capability across other EVQA datasets. Even though the accuracy on GQA-val and GQA-test is not the best, it still achieves sub-optimal results.

We also statistically assessed the significance of the difference between CACR and state-of-the-art algorithms using T-tests at the significance level $p = 0.05$. To equalize the contributions of different evaluation indicators to the overall variance, we use a method ranking metric for standardization. The T-test result between CACR and LININ is $p = 0.00079 < 0.05$, and the T-test result between CACR and VCIN is $p = 0.00002 < 0.05$. Both of them show statistically significant differences. These results indicate that CACR is effective in extracting reliable explanatory visual evidence from the image description.

As shown in Table 3, both automated metrics and human evaluations demonstrate that our proposed CACR achieves significant improvements in explanation consistency. Employing the same backbone network as VCIN, CACR elevates the consistency metric by 3.55%, with relative im-

Table 4

Ablation study of key CACR components on the GQA-REX validation set.

Model	BLEU-4	METEOR	CIDEr	GQA-test	OOD-test
w/o context generator	57.85	41.08	508.68	59.72	54.40
w/o L_{evi}	58.86	41.64	521.50	59.30	53.72
w/o L_{con}	59.06	41.74	523.12	59.66	54.54
Full	59.21	41.81	524.90	59.82	55.04

provements of 9.6%, 4.6%, and 10.1% on the Vis., Tex., and CoT. respectively. These results validate that our proposed CACR effectively establishes the causality between context and explanation, and enhances the consistency relationship between predicted answers and explanations, thereby strengthening the credibility of results.

5.6. Ablation study

We conducted an ablation study on the MLLM with and without prompt engineering, as illustrated in Fig. 5. Fig. 5(a) shows the unprompted result and Fig. 5(b) shows the prompted result. The results demonstrate that prompted descriptors are concise and contain minimal extraneous text. Compared with the unprompted results, the prompted results focus on the main object and its key actions, concisely explain the core relationships, and refine the key details to distinguish the objects. The prompted results significantly improve the information density by eliminating irrelevant information and using concise and efficient language, thereby effectively suppressing descriptive hallucinations.

To further isolate the contributions of the novel components introduced in CACR, we still conducted ablation studies on three key variants:

- w/o context generator: Removes the context generator; the pre-extracted MLLM descriptions are directly used as the visual context C during both training and inference, bypassing the learned generation process.
- w/o L_{evi} : Removes the evidence constraint loss that aligns the semantic distributions of the explanation and the context. Note that this variant corresponds to removing the causal path $C \rightarrow E$, as this path is implemented through the evidence constraint loss.
- w/o L_{con} : Removes the context generation loss; the context generator is trained without supervision from the MLLM descriptions.

Table 4 reports the results on the GQA-REX validation set. The results revealed several key findings. Firstly, removing the context generator led to a significant decline across several metrics (CIDEr -16.22, BLEU-4 -1.36) but the GQA-test accuracy drops only marginally (from 59.82 to 59.72). This discrepancy reveals a diagnostic signature of information leakage. The model can still predict correct answers by exploiting the high-level semantic shortcuts provided by the raw MLLM descriptions, but the generated explanations severely lack quality and grounding because the model bypassed learning proper visual feature extraction and multimodal alignment. This observation empirically validates the information leakage concern raised in Section 4.5. Secondly, removing L_{evi}

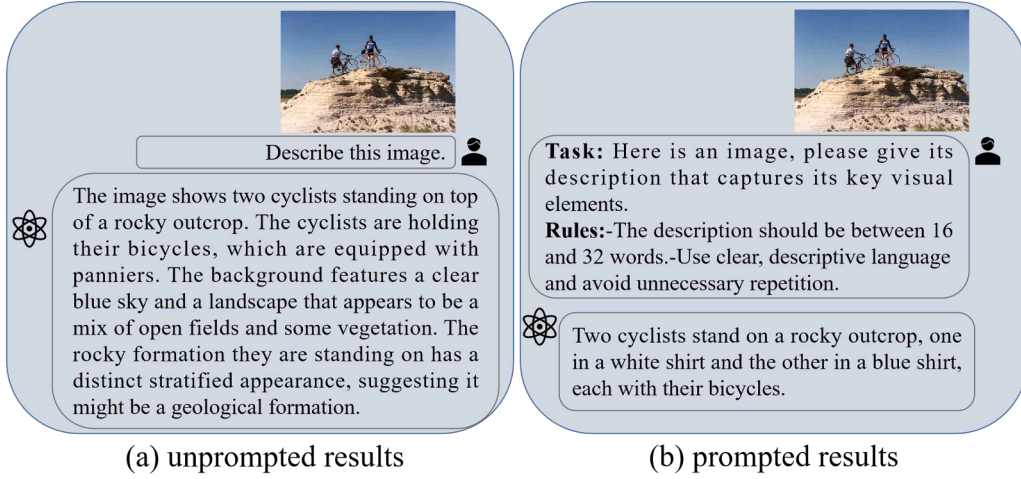


Fig. 5. MLLM ablation learning with/without prompts.

Table 5

Recall rates of key attributes related to visual skills for explanation generation.

Model	Color	Material	Sport	Shape	Pose	Size	Activity	Relation
REX-VisualBert	56.01	49.27	72.77	40.64	74.80	65.31	46.58	29.00
REX-LXMERT	65.38	60.22	70.16	51.95	74.41	69.83	45.75	29.83
LININ*	69.18	65.09	82.20	56.30	85.64	77.29	60.82	32.36
VCIN	69.38	76.67	81.67	67.82	85.37	77.28	61.64	80.43
CACR(Ours)	70.82	80.13	84.29	74.82	86.42	78.19	67.67	85.09

(i.e., the $C \rightarrow E$ path) resulted in a significant performance drop (CIDEr -3.40, OOD test -1.32), confirming that aligning the semantic of the explanation with the visual context is crucial for reducing explanatory hallucinations, especially in out-of-distribution scenarios. Thirdly, removing L_{con} only led to a relatively small decline (CIDEr -1.78), indicating that although the loss of context generation contributes to performance, its impact is not as significant as evidence constraints or the generator architecture itself. These results collectively validate the necessity and complementarity of each new component introduced in CACR.

5.7. Key attributes in explanation generation

To further evaluate the model's ability to recognize key visual attributes (i.e., color, material, sport, shape, pose, size, activity, and relation) in explanation generation, we compute recall rates on GQA-REX. As shown in Table 5, CACR significantly enhances the recall rates of various attributes. These gains can be attributed to the use of description, which reveals key evidence for explanation. To further interpret the effectiveness of description based context, we analyze the recall improvements across all eight attribute categories in Table 5. The largest gains are observed in Shape (+7.00), Activity (+6.03), and Relation (+4.66), indicating that descriptions provide critical global semantics for object shape recognition, action understanding, and inter-object relationships—tasks that are poorly supported by sparse region features. Moderate improvements appear in Sport (+2.09) and Material (+3.46), where descriptions offer supplementary cues. In contrast, gains on Color (+1.44), Pose (+0.78), and Size (+0.90) are marginal, suggesting that region features are already adequate for these perceptual judgments. This pattern confirms that the context mechanism is most beneficial for complex reasoning tasks that rely on global scene structure, while its advantage diminishes for simple attribute queries.

5.8. Analysis on MLLM parameters

To assess the impact of different MLLM parameters on our approach, we additionally employ the 72B version of Qwen2-VL (Wang et al.,

Table 6

Results of training CACR using visual contextual text generated by MLLM with different parameters.

MLLM	BLEU-4	METEOR	CIDEr	GQA-test	OOD-test
7B	59.21	41.81	524.90	59.82	55.04
72B	59.40	41.89	526.99	59.55	54.29

2024) to generate the visual contextual text data and train our model. As shown in Fig. 6, compared with Qwen2-VL-7B, the visual contextual text generated by Qwen2-VL-72B adopting the same prompt has richer semantic information, which reveals the characteristics or details of visual objects more effectively. However, as the experimental results represented in Table 6, compared to Qwen2-VL-7B, Qwen2-VL-72B (Wang et al., 2024) generates richer contextual information that facilitates more precise explanation generation, yet reduces question answering accuracy. This occurs because the expanded visual contextual text promotes consistency between context and explanation, thereby improving the quality of the explanation. But overly detailed visual descriptions introduce excessive visual information, making it impossible to simultaneously balance the accuracy of both the explanation and the answer. Moreover, in terms of metrics, the large model with tenfold parameters (whose quantitative version slightly underperforms the actual parameter count) yielded less than a 1% improvement in explainability. Therefore, through the intervention-based sensitivity analysis and qualitative analysis, we verified whether the description of CACR relies on MLLM generation.

5.9. Intervention-based sensitivity analysis

To analyze the sensitivity of CACR's explanation generation to the content of the visual context, we conduct intervention-based sensitivity analysis by systematically perturbing the generated context. Specifically, as shown in Fig. 7, we conducted two sets of intervention experiments: counterfactual context intervention and MLLM-generated description intervention. We achieve counterfactual intervention by

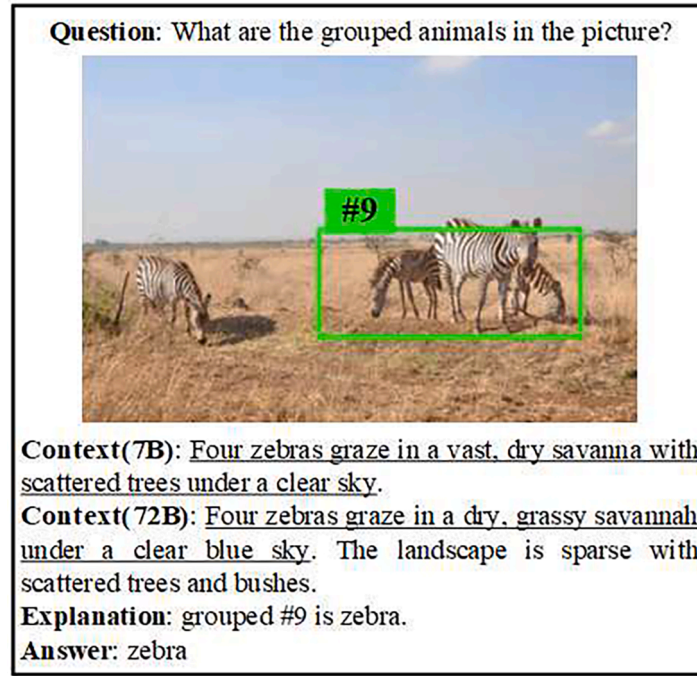


Fig. 6. Comparison of the image description generated by Qwen2-VL-7B and Qwen2-VL-72B, where **Context** represents the generated visual contextual text.

deleting and replacing visually relevant words and relational statements within the context generated by CACR, and achieve MLLM intervention by replacing the entire context generated by CACR with a model larger than the prompt MLLM.

We generated pre-extracted image description data via MLLM prompts to train the model to generate context, thereby enabling the model to derive reliable explanatory evidence from it. Consequently, we first applied counterfactual interventions on the Context text generated by CACR to analyze the sensitivity from C to E. As demonstrated by the counterfactual intervention sample Fig. 7(a), we removed the visual information ‘on the beach’ in Context that pertains to the question’s focus on ‘where’. CACR could only explain and answer the question by perceiving the semantic information of pure visual object #14. Thus, the weight assigned to ‘sand’ within the multi-semantic tokens associated with Object #14 has surpassed that of ‘beach’, which ought to be the more accurate term. Similarly, we replace ‘laptop’ in Fig. 7(b) with the synonym ‘computer’, and the corresponding explanations and answers will also change. In counterfactual example (c), we replaced the object of interest in the context with an erroneous object, which led the CACR to output incorrect explanations and answers. In counterfactual example (d), we removed the visual relational information pertinent to the question and reconfigured the remaining information into an erroneous visual logical relationship, directly causing the CACR to output explanations and answers with logically incoherent semantics. Counterfactual examples (a) and (b) in Fig. 7 demonstrate that CACR can enhance the quality of explanations and answers by leveraging contextual information. However, our negative intervention experiments in counterfactual examples (c) and (d) demonstrate that the quality of context is considerably important.

We conducted an MLLM intervention experiment to validate the impact of descriptions generated by MLLM at different levels on the CACR. The MLLM intervention employs the current strongest MLLM within the Qwen series, namely Qwen3-VL-235B-A22B (Yang et al., 2025), which generates descriptions to replace the CACR Context trained using pre-extracted description data from Qwen2-VL. From the results, it is evident that more powerful MLLMs generate more detailed contextual descriptions. However, overly detailed descriptions can still introduce noise, leading to a decline in explanation quality. As shown in Fig. 7, the

MLLM intervention sample (d) provides a more detailed and semantically specific description compared to the context generated by CACR. Nevertheless, the visual object recognition is lower than that of the pre-intervention explanation. This observation indicates that the granularity of the description is more likely to affect the quality of the generated explanations rather than the size of the model. This highlights the need for adaptive context filtering in future work.

5.10. Qualitative study

To present and analyze the results in an illustrative method, we conducted a qualitative study of the generated descriptions, explanations, and predicted answers. Fig. 8 shows four examples, with fair comparisons against the reproducible state-of-the-art model VCIN under the same VLPM framework and parameter optimizer. CACR outperforms VCIN in both explanation generation and answer prediction: (1) In case (a), in terms of explanation errors caused by ignoring key visual objects, due to the introduction of fine-grained descriptive information, CACR can generate more accurate explanations and predict answers. (2) In case (b), CACR is more resistant to explanatory illusions caused by visual logical errors, as demonstrated by VCIN’s misunderstanding of object #33 as bushes. (3) The GQA-REX dataset contains noisy explanations, such as the logically inconsistent explanation with the predicted answer in case (c). Thus, case (c) shows the excellent robustness of CACR explanations. Synthesizing the above three cases, we observe that the description-based context is most beneficial when the question requires relational reasoning or activity understanding (cases (a) and (b)), where the holistic description compensates for the sparsity of region features. These observations align well with the quantitative improvements on Relation (+4.66) and Activity (+6.03) reported in Table 5, confirming that the context mechanism effectively addresses explanatory illusions in complex visual reasoning scenarios. Meanwhile, case (c) demonstrates the model’s robustness against noisy ground-truth explanations, as the global context provides consistent semantic constraints that help maintain logical coherence. (4) CACR still has certain limitations. When the amount of key information in an image exceeds the length limit of our prompting strategy, the output text description will ignore visual relationships and only output the key detected object semantics, as shown

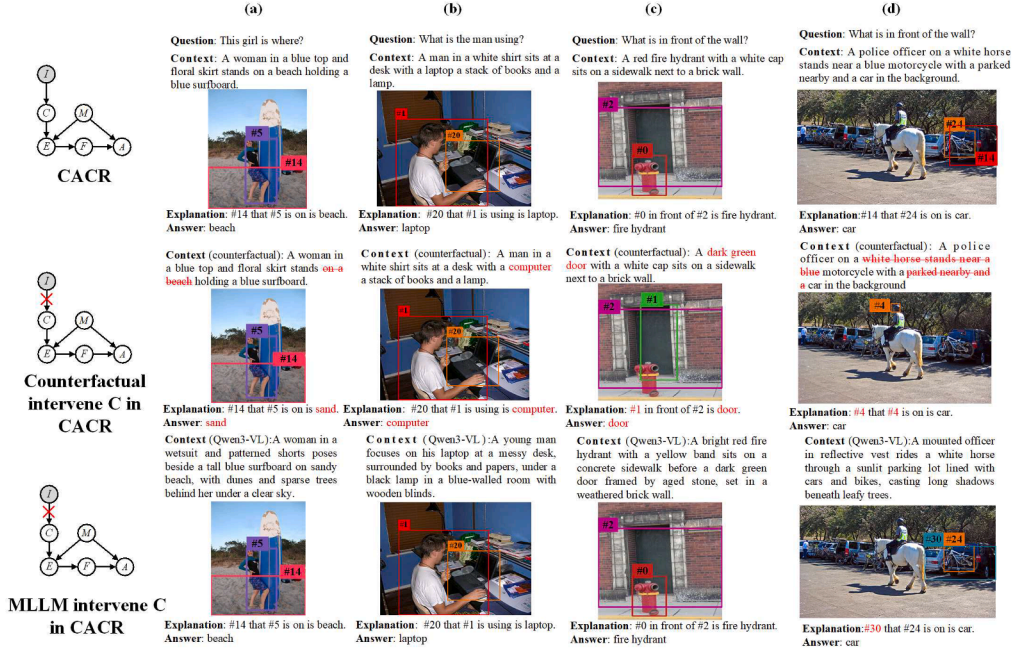


Fig. 7. Intervention-based sensitivity analysis for CACR, where red text in Context indicates the intervention as replacement text, red text with a strikethrough indicates the intervention as deletion text, and red text in explanations and answers indicates differing outcomes following the intervention.



Fig. 8. Qualitative study for EVQA, where **Context** represents the generated descriptions, red markings indicate incorrect relationships and answers, while green markings indicate correct, and blue markings indicate original noise data.

in example (d). Although CACR can obtain more accurate visual objects than VCIN in explanations, it still cannot generate correct explanations due to the lack of relational information. This failure case (d) reveals the critical bottleneck of description completeness: without sufficient relational evidence in the input context, even accurate visual region features cannot compensate for the omission. Such scenarios typically occur in highly cluttered scenes where the visual information density overwhelms the static prompting strategy. The evidence-constraint loss can mitigate noise but cannot recover missing relational semantics, as also evidenced by the counterfactual intervention in Fig. 7(d) where deleting relational information directly degrades explanation coherence.

Overall, CACR can effectively reduce the phenomenon of explanatory illusions where incorrect explanations yield correct answers. While current research lacks established quantitative metrics for measuring such illusions, the substantial body of visual qualitative results provides sufficient support for this conclusion. Moreover, combined with the findings of Section 5.9, the strength of the MLLM employed for pre-extracting descriptions does not necessarily correlate with higher description quality. Rather, truly high-quality descriptions are achieved through an adaptive balance between the visual information density of the image and the granularity of the contextual description. Enhancing the adaptive capability of the model will be the focus of our future work.

6. Conclusions and future work

To address the issue of explanatory illusion, we proposed the CACR model for EVQA in this paper. We first utilize the MLLM and design prompting strategies to generate high-quality descriptions. Then, we devise a new SCM to establish the causality among descriptions, explanations, and answers. After that, we convert the SCM into a deep variational inference network that employs the causal reasoning of description and multimodal explanation to predict the answer. Extensive experiments indicate the superiority of CACR in terms of credibility, robust generation ability that significantly reduces the risk of explanatory illusion. We note that while KL-based evidence constraints help promote semantic alignment between explanations and visual context, they do not independently guarantee visual faithfulness at the regional level. In fact, by reducing semantic drift, they represent a step toward achieving more faithful explanations, and supplementary evidence from anchoring metrics and intervention experiments collectively supports the credibility of CACR. Current limitations of CACR stem from potential noise in MLLM-generated contexts and insufficient information description for the MLLM prompting strategy. While CACR includes a context generator that learns to generate descriptions during training, its purpose is to obtain robust representations of visual context for causal reasoning, rather than to improve the quality of the input descriptions. Pre-extracted MLLM descriptions remain an upstream dependency that our framework cannot control. Thus, motivating our future research to focus on adaptive visual information context filtering methods that can find important visual evidence to generate more rationale explanations. We also note that the EVQA community currently lacks a unified benchmark that accommodates both region-level feature-based deep models and MLLM/agent-based approaches, as they differ fundamentally in output formats (such as visual token indices vs. natural language/bounding boxes) and evaluation protocols. Establishing such benchmark data and systematic testing standards is an urgent research requirement.

CRedit authorship contribution statement

Jiali Miao: Writing – original draft, Methodology, Conceptualization; **Xiaoling Huang:** Writing – review & editing, Methodology; **Kui Yu:** Writing – review & editing; **Yecheng Shi:** Visualization, Validation, Methodology; **Xiang Wang:** Writing – review & editing; **Fuyuan Cao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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